MILITARY SPECIFICATION
CONNECTOR ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of connector assembly used on T-4090 Test Set for M454 Projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 ----- Preservation, Methods of

STANDARDS

MILITARY

AN340-3 ------- Nut
AN515-3R4 ----- Screw
AN935-3 ------- Washer
MIL-STD-105 --- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 -- Lot Numbering of Ammunition

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the currentversion before use.

MIL-STD-1169 -- Packaging, Packing and Packing for Shipments of Inert Ammunition Components
NAS1201C6A ---- Coupling
NAS1201C6B ---- Coupling Offset
NAS1201C6-10 - Chain, Bead

DRAWINGS

ORDNANCE CORPS

8808362 ------- Packing and Marking for Storage and Shipments
8824556 ------- Connector, Electrical
8834564 ------- Connector Assembly
8883539 ------- Bag, Packing for Repair Parts
EL-8834564 ---- Equipment List

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The connector assembly shall comply with all requirements specified on the drawing (dwg) 8834564, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Circuit resistance. The circuit resistance shall comply with table I when tested as specified in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the current version before use.

Table I

Circuit Resistance (see dwg 8834564)

From Pin	To Pin	Ohms max
2	9	0.1
6	13	0.1

3.4 Insulation resistance. The insulation resistance of any circuit shall not be less than 20 megohms when subjected to a potential of 500 volts direct current (vdc) with the current limited to 5 milliamperes (ma) when tested as specified in 4.4.2.

3.5 Workmanship.

3.5.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.5.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.5.3 Burrs. Parts shall be free from burrs.

3.5.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.5.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.5.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges or points. All parts shall be free from flux, chips, dirt, grease rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the currentversion before use.

herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the currentversion before use.

Table II

Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:17Z
Check the source to verify that this is the current version before use.

4.3.2.1.1 Coupling (see NAS1201C6A); screw (see AN515-3R4); washer (see AN935-3); nut (see AN340-3); coupling offset (see 1201C6B); chain, bead (see NAS1201C6-10); connector, electrical (see dwg 8824556) covering details of dwg 8834564.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Objective evidence (see 6.2) for components in accordance with the applicable drawing or standard	Visual
Minor		
201.	Workmanship (see 3.5)	Visual/Manual

4.3.2.1.2 Connector assembly (see dwg 8834564).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Distance from flange to top of potting (.40)	Gage
102.	Length of potting (1.00)	Gage
103.	Potting flush or below flush	Visual
104.	Material (see 3.1)	Visual
105.	Potting completely covers wires	Visual
Minor		
201.	Workmanship (see 3.5)	Visual/Manual

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the current version before use.

4.3.2.1.3 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Preservation, packaging, packing and marking in accordance with MIL-STD-1169	Visual/Manual
Minor:	None defined	

4.3.2.1.4 Bag, packing for repair parts (see dwg 8883539) covering part number 8883539-2) and 5.2.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Material (see 3.1)	Visual
102.	Length before packing, min.	*Measure
103.	Width before packing, min.	*Measure
104.	Properly packed	Visual/Manual
105.	Proper heat seal	Visual/Manual
106.	Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual
107.	No tear, split, or hole in bag	Visual
Minor		
201.	Marking legible	Visual
202.	Workmanship (see 3.5)	Visual/Manual

*Measure with suitable scale.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the current version before use.

4.3.2.1.5 Packing and marking for storage and shipments (see dwg 8808362) and 5.2.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major		
101.	Gross load weight, max.	*Measure
102.	Contents tight in package	Manual
103.	Correctly packed	Visual/Manual
104.	Material (see 3.1)	Visual

*Measure with suitable scale.

Minor		
201.	Marking legible	Visual
202.	Workmanship (see 3.5)	Visual/Manual

4.3.3 Testing. The following tests shall be performed.

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major

These tests are not destructive. Samples tested may be returned to the lot.

4.3.3.1 Circuit resistance (see 3.3). Each connector assembly shall be tested in accordance with 4.4.1. If any item fails to comply with 3.3, the item shall be rejected and removed from the lot.

4.3.3.2 Insulation resistance (see 3.4). Each connector assembly shall be tested in accordance with 4.4.2. If any item fails to comply with 3.4, the item shall be rejected and removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:17Z
Check the source to verify that this is the current version before use.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL's) (See EL-8834564). When either of the conditions listed below exists, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance (see 3.3). The circuit resistance shall be measured in accordance with Table I (see 3.3) with approved measuring equipment.

4.4.2 Insulation resistance (see 3.4). The insulation resistance shall be measured with approved insulation resistance measuring equipment in accordance with Table IV.

Table IV

Insulation Resistance (see dwg 8834564)

Positive Potential to Pin	Negative Potential to all other Pins and Shell except
2	9
6	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:17Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The connector assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The connector assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The connector assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The connector assembly shall be packaged in accordance with dwgs 8883539 and 8808362, using the specified moisture barrier.

5.2.1.2 Level C. The connector assembly shall be packaged in accordance with dwg 8808362, except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The connector assembly shall be packed in accordance with dwgs 8883539 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwgs 8883539 and 8808362 as applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:17Z
Check the source to verify that this is the currentversion before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number and date of this specification.

b. Production or Repair Parts.

c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on DI-E-1118 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

Custodian:
Army - MU

Preparing Activity:
Army - MU

Project No. 5935-A 871

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:17Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:17Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-48197(MU) Connector Assembly

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING:

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T19:17Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:17Z
Check the source to verify that this is the current version before use.